

Python Web Frameworks

Introduction

Web frameworks are software libraries designed to support the development of web applications, including web services, web resources, and web APIs. They handle common tasks such as routing, templating, and authentication. Choosing a framework early in a project has long-term architectural impact.

Popular Frameworks

Flask

Flask is a lightweight WSGI web application framework designed to make getting started quick and easy, scaling up to complex applications.

Django

Django is a high-level Python web framework that encourages rapid development and clean, pragmatic design built by experienced developers.

Comparison

Framework	Type	First Release
Flask	Micro	2010
Django	Full	2005
FastAPI	Async	2018

Key Features

- URL routing and request dispatching
- Template rendering engine
- Database abstraction layer
- Built-in authentication and authorization

Code Example

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def hello():
    return "Hello, World!"
```

Conclusion

Choosing the right web framework depends on your project requirements and team expertise. Each framework has its own philosophy and trade-offs.